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A comprehensive data mining approach to estimate the rate of penetration: Application of neural network, rule based models and feature ranking

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ABSTRACT

Rate of Penetration (ROP) estimation is one of the main factors in drilling optimization and minimizing the operation costs. However, ROP depends on many parameters which make its prediction a complex problem. In the presented study, a novel and reliable computational approach for prediction of ROP is proposed. Firstly, *fscaret* package in R environment was implemented to find out the importance and ranking of the inputs parameters. According to the feature ranking technique, weight on bit and mud weight had the highest impact on ROP based on their ranges within this dataset. Also, for developing further models Cubist method was applied to reduce the input vector from 13 to 6 and 4. Then, Random Forest (RF) and Monotone Multi-Layer Perceptron (MON-MLP) models were applied to predict ROP. The goodness of fit for all models were measured by RMSE and R^2 in 10-fold cross validation scheme, and both models showed a reliable accuracy. In order to gain a deeper understanding of the relationships between input parameters and ROP, MON-MLP model with 6 inputs was used to check the effect of weight on bit, mud weight and viscosity. Finally, RF model with 4 variables was used to extract the most important rules from dataset as a transparent model.

1. Introduction

Oil and gas well drilling is one of the most important and expensive part of the oil industry. One of the main factors which has an effect on drilling efficiency and cost is the Rate of Penetration (ROP). It is important to find and optimize the relationship between drilling parameters and ROP in order to reduce cost and increase drilling operation's performance eventually. The ROP can be explained by the progress of a bit in rocks and formations in time units, and feet per minute and/or meters per hour are the commonly used units for that. There are several parameters have an effect on ROP such as bit type, Weight on Bit (WOB), revolution per minute (rpm), mud properties, drilling hydraulic, formation conditions, etc (Walker et al., 1986). Generally, ROP is surveyed in instantaneous and average types. Instantaneous ROP is recorded in restricted time and distance during drilling, like the data that is used in this work, while average ROP is recorded in time periods that drill pipes run in a well.

In the past, several researchers tried to develop models and formulate related parameters to ROP (Bourgoyne and Young, 1974; Warren, 1987; Winters et al., 1987); however, none of them could make an accurate and comprehensive estimation because of the complex and non-linear behavior of parameters with ROP (Ricardo et al., 2007). These difficulties have conducted the recent research toward using intelligent solutions which implement computers as an alternative (Rahimzadeh et al., 2010; Monazami et al., 2012; Edalatkhah et al., 2010).

This study is intended to propose a procedure for predicting the ROP and/or other complex systems in order to reach reliable models. It consists of using feature ranking technique and applying different data mining algorithms to search available design space and build predictive models.

2. Materials and methods

In this work, different computational intelligence methods have been used to model the ROP employing on a dataset containing 226 data points. These data points were collected from 5 different wells of a field located in South-West of Iran using the rotary table, and includes different characteristics of drilling operations, drilling hydraulic, etc. in paying and non-paying zones.

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Table 1
Input variables in feature ranking.

Variable	Range
WOB, klb	6–30
rpm	4–100
Flow Rate, gpm	350–1100
Pump Pressure, psi	1200–3800
Incline	5.1–56
Azimuth	52.41–322
Measured Drilling Depth, m	1020–5186
MW[a], ppg	8.85–17.25
Funnel Viscosity, sec	28–179
Plastic Viscosity, cp	3–97
Yield Point, lb/100sf	4–76
Gel 10s[b], lb/100sf	1–73
Gel 10m[c], lb/100sf	2–143

[a] Mud weight.
[b] Gel strength after 10 s.
[c] Gel strength after 10 min.

2.1. Data mining

Using new technologies in the present era, causes a large volume of data to be created in a rapid pace, thus our success in data interpretation depends on a comprehensive insight within it. Emerging advanced processing systems over the last decade, has allowed us to put away the old time-consuming and weary data analysis methods. Data mining is the term that is used for processes of finding hidden patterns and correlations through data to predict the outcomes, and is comprised of three main scientific fields of statistics, artificial intelligence and machine learning. Statistics and Artificial Intelligence (AI) can be found extensively in the literature (Freedman, 2009; Radermacher, 1991). In machine learning, the computers are used to probe structures within a dataset even if there is not a theory behind the way they look like (Kordon, 2010). In this work, different machine learning approaches have been applied to model the ROP based on various input variables.

2.2. Feature ranking

Dealing with many features in machine learning problems, can cause some difficulties in both the dataset and the derived models. Feature ranking is a technique which finds the importance of features, and can be used to reduce the dimensions of the dataset. Using this technique also has the benefits of shorter training time, ease of interpretation of models, overfitting reduction and lower cost in data collection (James et al., 2013). Depends on the choice of predictive models for feature ranking approach, different results of variable ranking and importance would be achieved (Szlek et al., 2016). Though, it is always a moot point to choose a proper model in feature ranking.

In this work, *fscaret* package as a in R environment is used to reduce the input vector and give stable variable importance (Szlek and Mendyk, 2015; Team, 2015). *fscaret* uses a large number of 103 various models in order to determine the importance of features and rank them. At the end, the importance of variables is averaged through all models to give reliable results, and is scaled from 0 to 100.

In this work, firstly a total number of 21 variables were chosen to estimate the output. Then, 8 constant or nearly constant variables were excluded, and 13 remaining variables were selected for feature ranking. These features are shown in Table 1.

2.3. Model assessment

To evaluate the model's goodness of fit, Root Mean Squared Error (RMSE) and the coefficient of determination (R^2) was used. Additionally, to examine the generalization ability of the models, 10-fold Cross Validation (10CV) approach was performed. The model which has a lower RMSE and higher R^2, can be considered as the best model. RMSE and R^2 are calculated based on Equations (1) and (2).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}\left(y_{prd,i} - y_{act,i}\right)^2}{n}} \tag{1}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_{prd,i} - y_{act,i}\right)}{\sum_{i=1}^{n}\left(y_{prd,i} - y_m\right)} \tag{2}$$

where y_{act} and y_{prd} are the actual and predicted values; respectively, i is the data record number, y_m is average of the experimental value, and n is the total number of records.

It is essential to check the generalization ability of produced models in data mining problems. One of the common methods for model's validation is to split the dataset into training and test sets, and evaluate the performance based on the test set. However, inability of measuring the generalization ability of models is a weak point of this method, and it is possible that accurate models are generated by chance especially in low to medium sized datasets.

One of the techniques to survey how a model can generalize to an independent data set is 10CV (Zhang and Yang, 2015; Bahrami et al., 2016). The data set is split into 10 parts where; at first step, one of the 10 subsets is chosen as the test set, and the other 9 subsets are all considered as the training set. This procedure continues 10 times, and at each step one of the parts is recognized as the test set. Then, the average of RMSE and R^2 is calculated for all 10 trials. 10CV has a lower variance compared to other methods, and can confirm that the model is generalized and not over-fitted. Fig. 1 demonstrates the 10CV procedure.

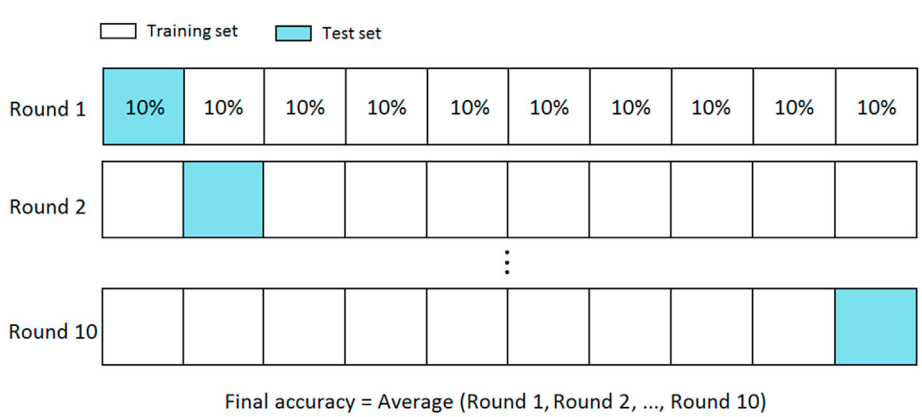

Fig. 1. Schematic of 10CV method.

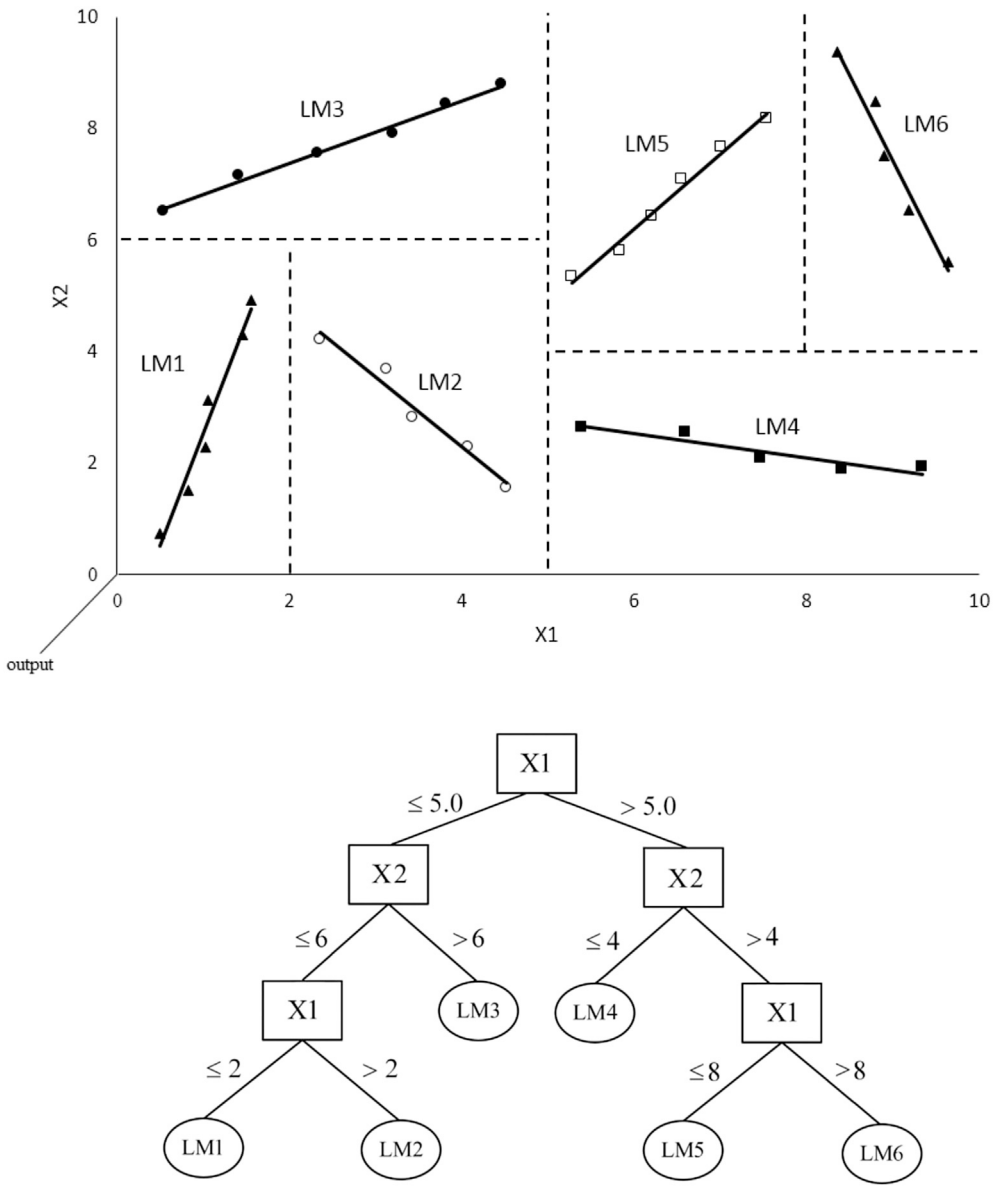

Fig. 2. Example of one Cubist tree with 6 linear regression models.

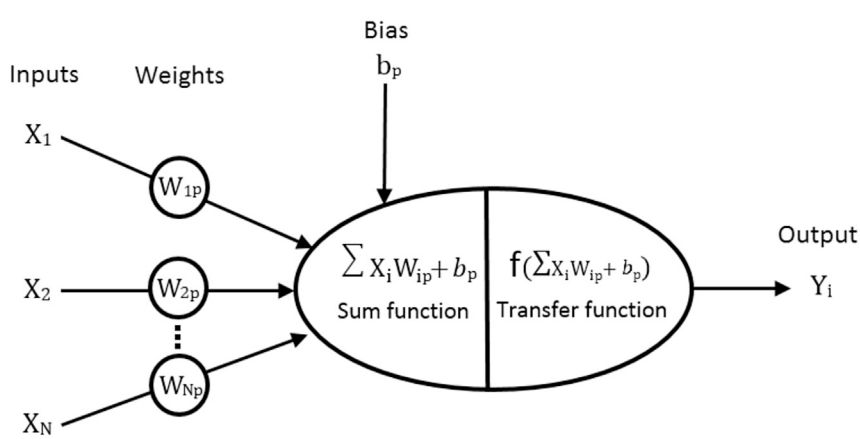

Fig. 3. Scheme of a single neuron.

2.4. Decision trees

One of the fast and efficient methods for data modeling is decision tree learning which is suitable for high dimensional data. In these tree-structured models, construction starts at root node, and it splits into children nodes. Splitting of nodes continues until they end at leafs. Each interior node corresponds to an attribute, each leaf node represents a value of the output variable, and each branch represents conjunctions of features that lead to those values. Splitting of training set to smaller subsets recursively are done based on greedy algorithm (Quinlan, 1986), and each node is evaluated by an Impurity Function which decreases as new nodes are generated (Breiman et al., 2000). Children nodes are more homogeneous, and have lower impurity compared to their parents' nodes. The splitting stops when the impurity of children nodes doesn't decrease, and finally a tree is formed.

Although decision trees are easy to interpret, they have some disadvantages which limit their usage. Overfitting is one of the most important issues that reduce the accuracy of models especially in larger trees. Also, a small perturbation in training data can result in severe changes in obtained trees. To overcome the problems associated with decision tree models, it is offered to use ensembles of trees such as Random Forest (RF) and Cubist.

2.4.1. Random forest

Breiman in 2001 firstly introduced RF. It is an ensemble of many unpruned decision trees which are sequentially grown, instead of only one. RF applies bootstrap sampling technique that randomly selects a group of dataset with replacement, and uses the remaining samples for testing each tree. This process repeats for all trees, and random sampling ensures the diversity among the ensemble of trees and enhances the predictions. At the end, an internal cross-validation makes an average of errors within all trees.

This method randomly picks a small subsets of size m out of M features for each node to train a tree ($m < M$). Size of m is kept constant during the forest growing, but the features are changed for each node. As a result, the best split for each node is chosen among m features rather than all features of M.

There are different parameters which RF uses to grow the trees: number of trees in forest (ntree), size of subset (mtry) and maximum number of nodes in trees (maxnode). In practice, ntree and mtry should

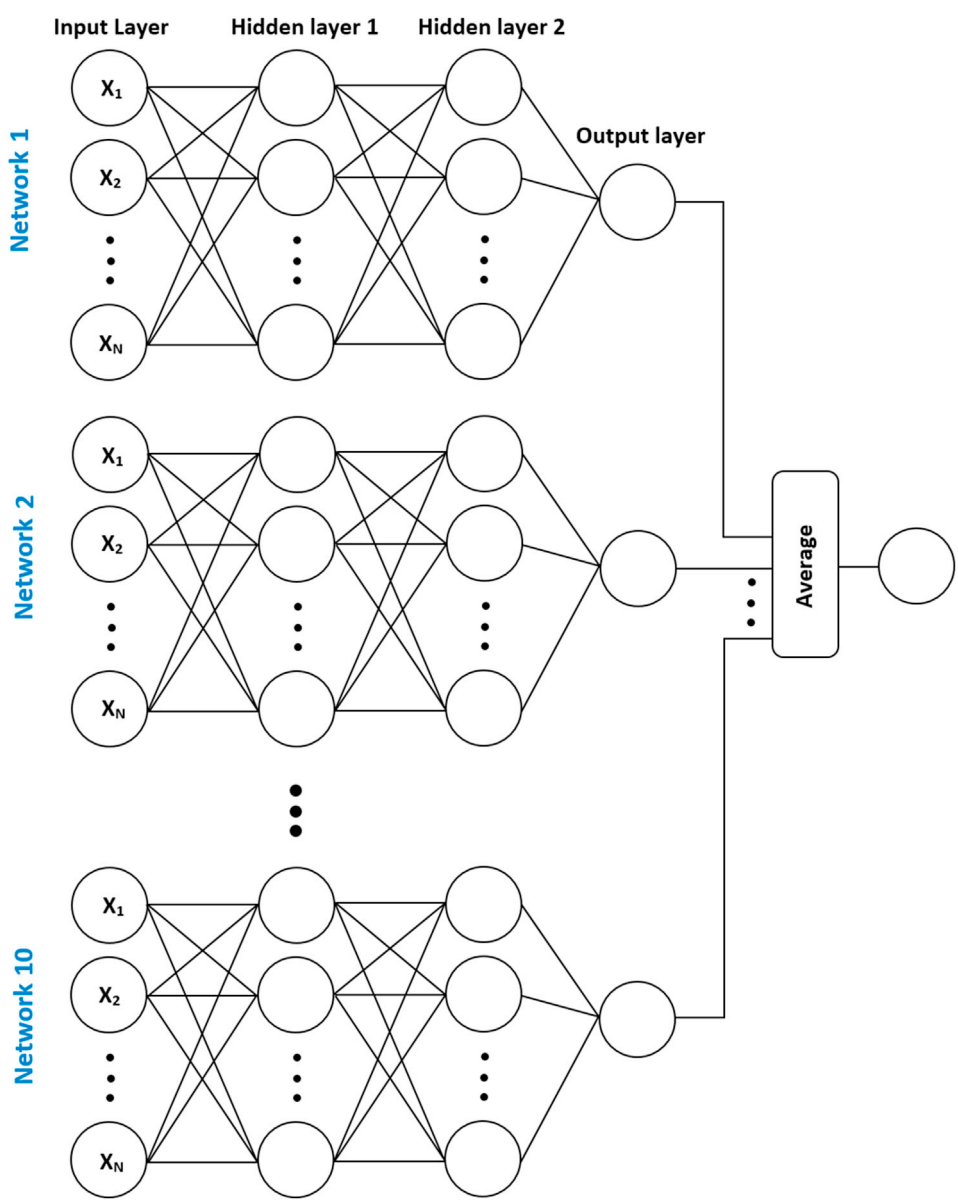

Fig. 4. Scheme of MON-MLP model.

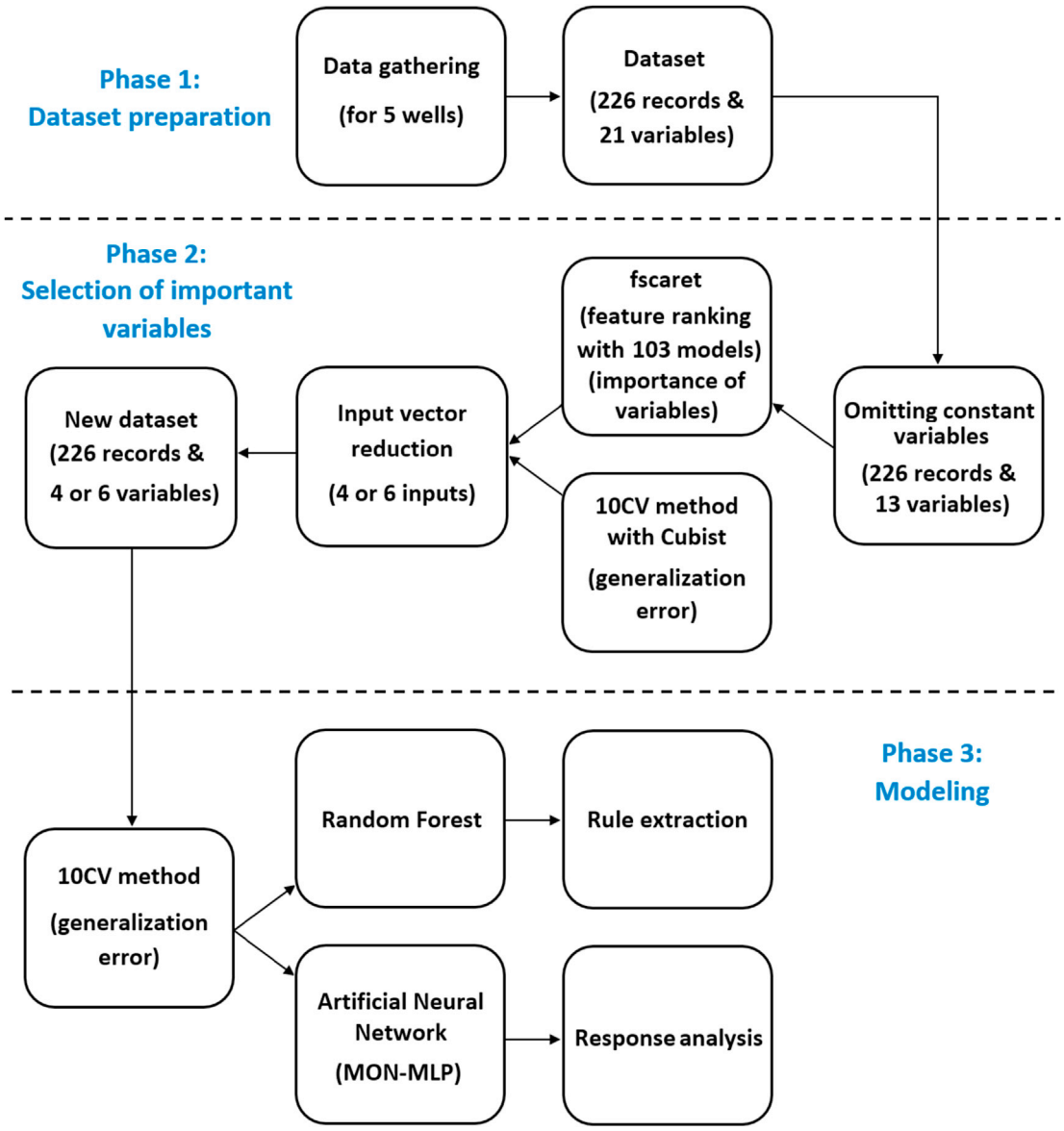

Fig. 5. General Scheme of workflow.

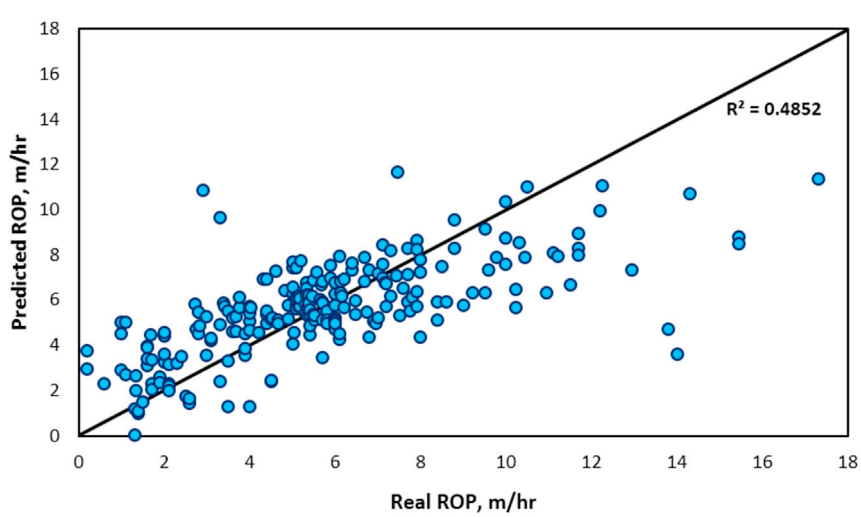

Fig. 6. Linear regression on the dataset.

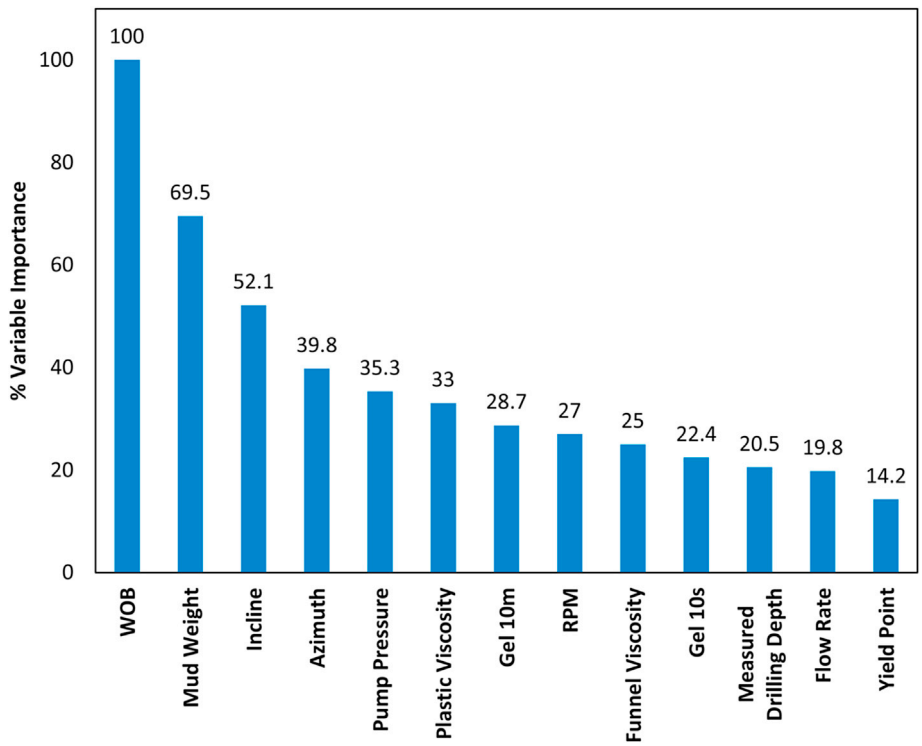

Fig. 7. Importance of variables obtained from fscaret.

Table 2
Controlling parameters for RF models.

Controlling Parameters	RF (4 variables)	RF (6 variables)
mtry	4	6
ntree	20	20
maxnode	200	800

be optimized to find the best accuracy in ensembles of trees. When features are replaced by random noise, the importance of each variable also can be quantified by calculating the model accuracy. Hence, importance determination of features is another advantage of implementing RF.

2.4.2. Cubist

Cubist is a rule-based predictive model which was first introduced by Quinlan and is an extension of earlier M5 model (Quinlan, 1992), and it constructs a type of regression tree where the prediction is based on linear regressions, not discrete values, of terminal nodes. The final model of Cubist is a set of rules (paths from root to leaves), where each rule corresponds to a multivariate linear expression. In other words, if all conditions within a rule are met, then the linear expression can be applied to predict the target value. Fig. 2 shows an example of one Cubist tree with 6 Linear Models (LM).

Cubist reduces the rules via pruning and/or combining with nearest neighbor rules in order to decrease the absolute error and to stop overlapping the rules (Quinlan, 1992). This pruning/combining procedure makes the model more robust to outliers and prevents it from overfitting.

2.4.3. MON-MLP

Artificial Neural Network (ANN) is one of the methods for data processing. Generally, ANN consists of three main layers, called input, hidden and output layers. There are some controlling parameters in a routine ANN which should be optimized to build a precise predictive model, such as; number of hidden layers, neurons of each layer, transfer functions, etc (Bashiri and Farshbaf Geranmayeh, 2011). Hence, a challenging step in ANN modeling is tuning of theses parameters, which is mostly done by try and error technique. In ANN, arriving signals (inputs) to each neuron in hidden layers, are multiplied by the adjusted weight elements (W_{ip}), combined, and then flow forward through a transfer function to produce the outputs for neurons. Fig. 3 illustrates a scheme of a single neuron in ANN modeling. The common transfer function which are mostly

Table 3
Summary of fitness error for all models.

Model	RMSE (Training)	R^2 (Training)	RMSE (10 CV)	R^2 (10 CV)
Cubist (4 variables)	1.4777	0.7787	1.7777	0.6457
Cubist (6 variables)	1.4548	0.7865	1.7987	0.6382
Cubist (all variables)	1.1444	0.8775	2.0134	0.5351
RF (4 variables)	0.7496	0.946	1.4718	0.7653
RF (6 variables)	0.6784	0.9572	1.5039	0.751
MON-MLP (4 variables)	1.0333	0.8871	1.4468	0.7618
MON-MLP (6 variables)	0.7005	0.9472	1.33	0.8009

Table 4
General rules extracted from RF with 4 input variables.

Rule	length	error	condition	prediction	Importance
1	3	0.328	12.5 < WOB & Azimuth ≤280.3 & MW ≤ 10.598,15	Medium	1
2	2	0.333	14.5 < WOB & 233.6 < Azimuth	Medium	0.509
3	5	0.2	WOB ≤12.5 & 33.5 < Incline & 71.2 < Azimuth ≤206 & MW ≤ 11.81	Medium	0.366
4	3	0	13.5 < WOB & Incline ≤48.65 & 280.3 < Azimuth	Medium	0.277
5	3	0.103	Incline ≤41.5 & 196.6 < Azimuth ≤225	Low	0.125
6	2	0.222	18.5 < WOB & 44.5 < Incline	High	0.118
7	1	0	55 < Incline	High	0.101
8	2	0.1	42.5 < Incline ≤44.1	Medium	0.089
9	3	0	WOB ≤12.5 & 9.89 < MW ≤ 11.47	Low	0.084
10	2	0	Incline ≤36.5 & Azimuth ≤190.6	Low	0.06
11	3	0	15.5 < WOB & 224.7 < Azimuth & MW ≤ 10.222,625	Medium	0.048
12	3	0.286	14.5 < WOB & 45.8 < Incline & 10.22 < MW	High	0.047
13	3	0	18.5 < WOB & 44.5 < Incline & Azimuth ≤224.7	High	0.045
14	4	0.444	14.5 < WOB ≤22.5 & 45.5 < Incline ≤55	Medium	0.029
15	2	0.25	20 < WOB & Incline ≤42.5	Low	0.026
16	2	0.135	Incline ≤43.5 & 10.14 < MW	Low	0.022
17	4	0.125	12.5 < WOB & 63.7 < Azimuth ≤73 & 9.39 < MW	Medium	0.015

employed in ANN are: hyperbolic tangent sigmoid (TANSIG), logarithm sigmoid (LOGSIG) and pure linear (PURELIN), and they are presented in Equations (4)–(6).

$$\mathrm{TANSIG} = \frac{2}{1+\exp(n_p)} - 1 \tag{4}$$

$$\mathrm{LOGSIG} = \frac{1}{1+\exp(-n_p)} \tag{5}$$

$$\mathrm{PURELIN} = n_p \tag{6}$$

$$n_p = \sum_{i=1}^{N} X_i * W_{ip} + b_p \tag{7}$$

where; X_i is the inputs, b_p is the bias and n_p is the transfer function input.

In this study, "monmlp.fit" function of the "monmlp" package in R environment was used to implement Monotone Multi-Layer Perceptron (MON-MLP) neural network (Cannon, 2014). This function includes an ensemble of MLP models, and it is used bootstrap aggregation (bagging) to help avoid overfitting. The final result was an average of RSME and R^2 of the ensemble of models. All of the generated models had two hidden layers with 2–20 nodes per layer. The transfer functions of TANSIG and PURELIN were used for hidden and output layers; respectively. Fig. 4 illustrates a schematic of MON-MLP model. Also, other parameters were as follows:

- The ensemble was a group of 10 models.
- Iteration numbers for training the models were set to 10, 50, 80, 100, 500, 1000 and 2000.
- "trial" function as a multi-start technique was set to 5 to avoid local minima.

The workflow of this study along with the results of feature ranking is demonstrated in Fig. 5 which consists of 3 main phases of data preparation, selection of important variables and modeling.

3. Result and discussion

As an initial effort, linear regression was applied to verify the complexity of the system and to find out whether machine learning approaches can improve the predictions. Fig. 6 shows the estimated results based on the linear regression model. From this figure, it can be concluded that the behavior of the system is highly complex ($R^2 = 0.482$), and confirms that further investigations are needed.

3.1. Feature ranking

The initial input vector consisted of 13 independent variables. To perform feature ranking and finding the importance of each variable, *fscaret* package in R environment was used. Fig. 7 demonstrates the results of feature ranking created by *fscaret*.

The selected variables were mostly consistent with the findings in literature. According to the results, WOB and MW were the first and second most effective parameters on ROP for this dataset, respectively. These two parameters were investigated in some literature as the main factors affecting the ROP (Akpabio et al., 2015; Ernst et al., 2007; Cheatham and Nahm, 1985). The parameters of incline and azimuth, which are related to the well trajectory, were the next important parameters for this formation type. The importance of incline and azimuth may differ for other formation types, and have a neglect effect on ROP. It should be mentioned that the result for variable importance is case specific, and for other reservoirs and formations, it may yield different results.

3.2. Machine learning approaches

3.2.1. Cubist

Cubist model which is a fast algorithm, was implemented on the dataset to find the best input vector's size based on the generalization error (RMSE). Hence, the models were trained on 10CV at various number of variables from the most important to the lowest one based on the results of feature ranking. According to the results, RMSE reduced from 2.0134 (for all variables) to the lowest amount of 1.7777 and

Table 5
Controlling parameters for MON-MLP models.

Controlling Parameters	MON-MLP (4 variables)	MON-MLP (6 variables)
Number of nodes in layer 1	12	20
Number of nodes in layer 2	6	2
Number of models in ensemble	10	10
Trials	5	5
Maximum number of iterations	90	200

1.7987 (for 4 and 6 variables). Finally, input vector size of 4 and 6 were chosen for further analyses to find more accurate results with other methods.

3.2.2. Random forest

To find the best RF model, around 3 000 models were trained with different controlling parameters (mtry, ntree and maxnode), and finally the ones which presented in Table 2 were chosen as the most accurate model.

To evaluate the RF model's performance, R^2 and RMSE based on 10CV approach was used. A summary of the results about all models with 4 and 6 input vector size is presented in Table 3. It can be concluded from this table that the models generated by RF (4 and 6 variables), were well trained, and were more accurate than the ones obtained from other models but they had a higher generalization error based on 10CV compared to MON-MLP models.

Rule extraction is one of the benefits of using RF models in comparison to black-box models such as ANN. It means that the generated trees can be expressed as some general rules which explain the relationships in the dataset, and are very easy to interpret. Therefore, "inTrees" package in R environment was applied on the ensemble of trees to extract the rules according to their length, error and importance (Deng, 2014). The extracted rules for RF model with the input vector size of 4 are shown in Table 4. The output (ROP) was automatically split into three levels, Low (0.183–5.9), Medium (5.9–11.6) and High (11.6–17.3) based on the minimum and maximum of it.

3.2.3. MON-MLP

It was concluded from Table 3, that the MON-MLP model with 6 variables was the best model in terms of generalization error (R^2 and RMSE in 10CV). So, this model was selected for further analysis to see the behavior of important variables on ROP. Similar to the RF procedure in finding the most accurate model, around 3000 models were trained to select the best MON-MLP model. A summary of these controlling parameters is presented in Table 5. Also, Fig. 8a and b demonstrate the predictions of ROP obtained from training and 10CV models with MON-MLP.

A survey was applied on parameters of WOB, MW and Plastic viscosity to see how they affect the ROP. Fig. 9a–c represent the effect of these inputs on ROP. To plot these figures, the variable under study was varied while the other variables were fixed at their midrange values. By analyzing Fig. 9a, it can be seen that up to a limit (flounder point), increasing the WOB enhanced the penetration. If too much weight is applied, ROP decreases, and may cause rapid bit wearing. Furthermore, an increase in MW generally tends to decrease the ROP; however, a slight growth in ROP was seen at low MW in Fig. 9b. This behavior may happen because of the positive effect of MW on hydraulic impact force which improves the ROP (Bourgoyne et al., 1986). At low plastic viscosities, increasing the viscosity improves the transport ability of cuttings, and therefore raises the ROP. But, excess amount of viscosity, inverses the

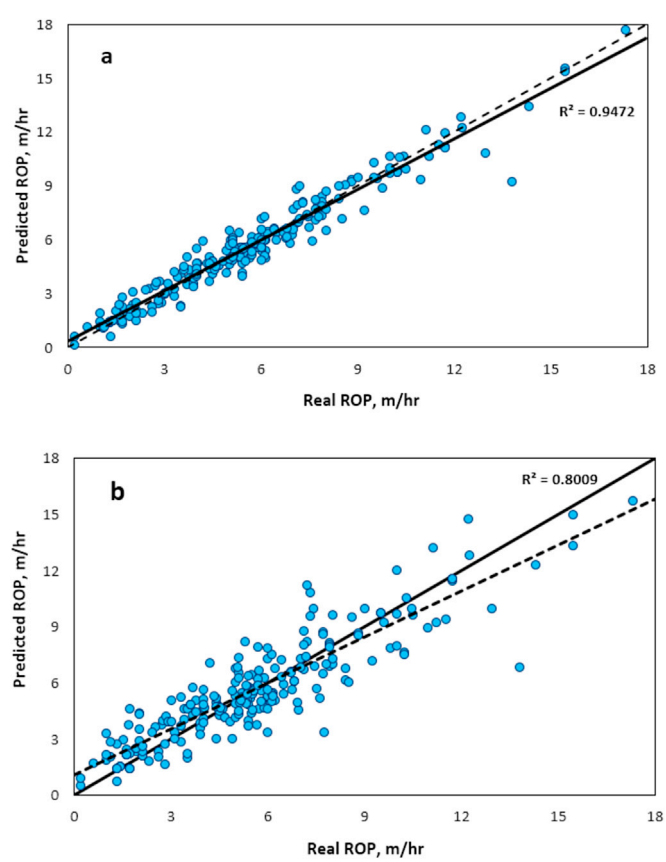

Fig. 8. Predicted values vs. actual values for a) training and b) 10-CV model of MON-MLP.

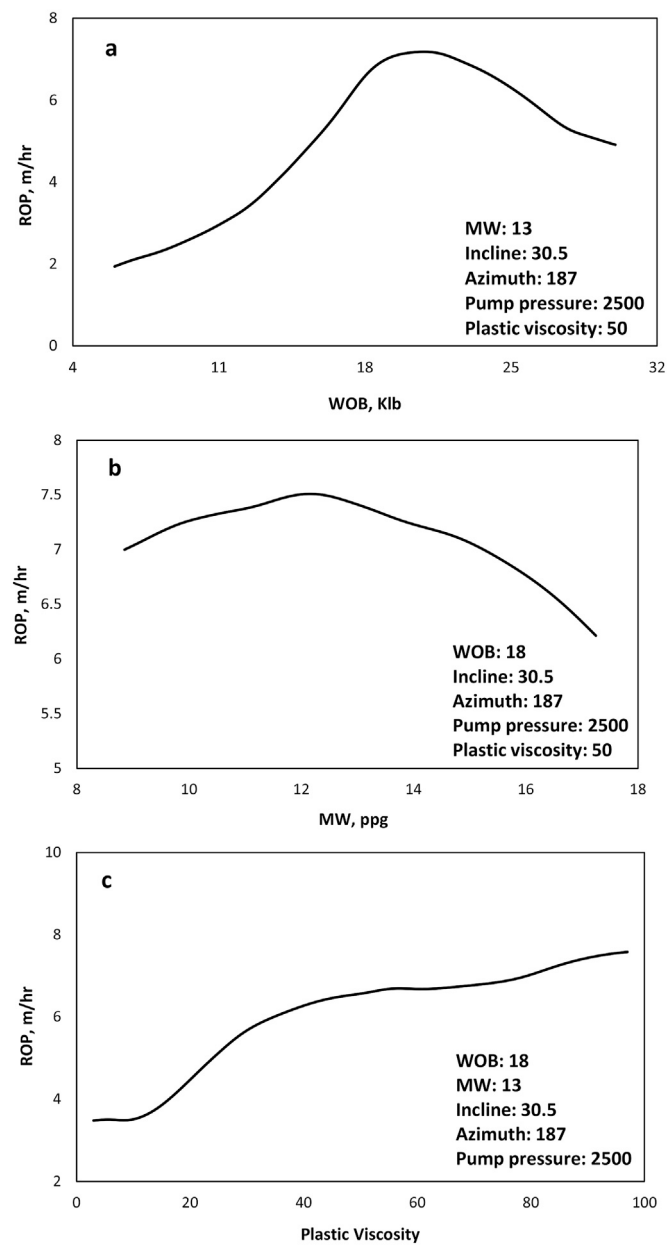

Fig. 9. a–c: Effect of WOB, MW, and plastic viscosity on ROP.

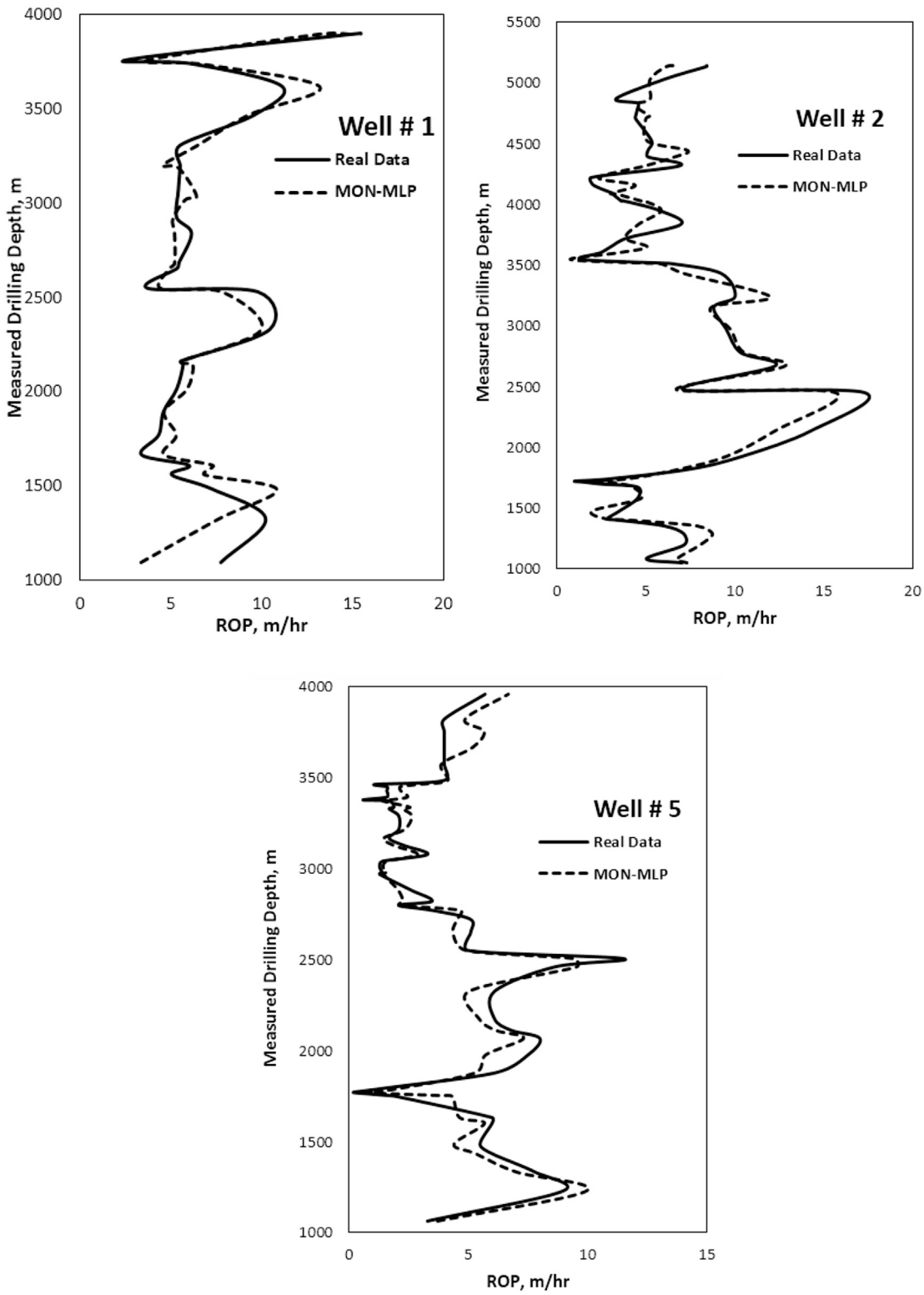

Fig. 10. a–c: Predicted and Actual ROPs at different depths for well 1, 2 and 5.

result. In Fig. 9c, because the viscosity range is limited, the inverse behavior didn't observe.

At the final effort, the predicted ROP were compared to actual ROP at various depth using MON-MLP model. Fig. 10a–c shows the predicted and actual ROPs vs. depth for well number 1, 2 and 5. As it can be seen, there is a reasonable match between the ROPs at various depth which implies the success of modeling with MON-MLP and the whole procedure done in this work. Therefore, the achieved MON-MLP model can be used as a reliable method for estimation of ROP depend on various parameters, and further analysis within the reservoir.

4. Conclusion

In this work, a comprehensive data mining approach is presented to perform the computational modeling of ROP. Moreover, following this approach and using the data mining techniques can help to unfold knowledge from the dataset. Applying *fscaret* package in R environment enabled to find the importance and ranking of all 13 variables. According to the results of feature ranking, WOB and MW had the most impact on the ROP. Furthermore, Cubist model was implemented to reduce the input vector to 4 and 6 which had the lowest RMSE of 1.7777 and 1.7987, respectively. RF and MON-MLP models were then developed on the dataset using the reduced input vector. Both models, showed a reasonable RMSE and R^2 based on 10CV approach. RF model as a transparent model, unlike other black-box ones, was used to extract the general rules from the dataset. So, the RF model with 4 variables (R^2 of 0.946 and 0.7653 for training and test models) was employed, and 17 general rules were extracted. These rules can help to better understand the relationships of features on ROP. Finally, the MON-MLP model with 6 variables which was the most accurate model among all (R^2 of 0.9472 and 0.8009 for training and test models) was used to predict the ROP at various depths for 3 wells. Also, the effect of WOB, MW and viscosity on ROP were investigated to see how they are behaving within the dataset.

Nomenclature

Abbreviations

10CV	10-fold Cross Validation
AI	Artificial Intelligence
ANN	Artificial Neural Network
Gel 10 s	Gel Strength after 10 s
Gel 10 m	Gel Strength after 10 min
LM	Linear Model
LOGSIG	Logarithm Sigmoid Transfer Function
maxnode	maximum number of nodes in trees
MW	Mud Weight
mtry	size of subset randomly sampled
MON-MLP	Monotone Multi-Layer Perceptron
ntree	number of trees in forest
PURELIN	Pure linear Transfer Function
R^2	coefficient of determination
RF	Random Forest
RMSE	Root Mean Squared Error
ROP	Rate of Penetration
rpm	revolution per minute
TANSIG	Hyperbolic Tangent Sigmoid Transfer Function
WOB	Weight on Bit

Symbols

b_p	ANN bias
m	random subset size
M	number of all parameters
n	total number of records
N	number of ANN inputs
n_p	transfer function input
p	number of ANN layers
W_{ip}	adjusted weight of input i for layer p in ANN
X_i	neurons input
y_{act}	actual value
Y_i	output of neurons
y_m	actual values average
y_{pred}	predicted value

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